PGPM · SYSTEMS SPECIALISATION

UNIT 01

Introduction to AI and Generative AI

What artificial intelligence really is, the different types, how it learns, and where generative AI fits in.

What is AI | Types & Models | Gen AI vs Predictive AI | Risks & Benefits

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AGENDA

What we will cover

Eight short stops. By the end you should be able to explain AI to someone with no technical background.

01

What Is Artificial Intelligence?

Starting with the basics - in normal language.

DEFINITION

Artificial Intelligence

Software that learns from data to do jobs that used to need a human brain.

- It looks at large amounts of past data
- It finds patterns in that data
- It uses those patterns on something new
- It gets better as it sees more examples
- No one writes step-by-step rules for it

Traditional software

A person writes the rules. Same input always gives the same output.

Artificial intelligence

The machine works out the rules from examples it has been shown.

Why it matters

You can now automate judgement, not just repetitive clicking.

EVERYDAY EXAMPLES

You already use AI several times a day

AI is not new or futuristic. It is already normal.

Maps and navigation

Predicts traffic and picks the fastest route using live and historic data.

Streaming and shopping

Suggests the next show or product based on what people like you chose.

Bank fraud alerts

Spots a payment that does not match your normal spending and blocks it.

Email spam filter

Learns from millions of flagged emails what junk looks like.

Face unlock

Matches the camera image to the stored pattern of your face.

Chat assistants

Understands your question in plain language and writes an answer.

02

Types of AI

Two useful ways to sort it.

TYPE 1 — BY CAPABILITY

How smart is it?

Almost every product marketed as 'AI' today is narrow AI.

1

Narrow AI

Very good at one job only. A chess engine cannot drive a car.

Status: this is everything we use today.

2

General AI

Would handle any task a human can, and switch between them.

Status: does not exist yet.

3

Super AI

Would beat humans at everything, including creativity.

Status: theory and debate only.

TYPE 2 — BY HOW IT WORKS

Four families you will hear named

They overlap - deep learning is a branch of machine learning.

Machine Learning

Learns patterns from structured data. Used for scores, forecasts and risk ratings.

Deep Learning

Uses layered neural networks. Handles images, audio and language well.

Natural Language Processing

Reads and writes human language. Powers chatbots and translation.

Computer Vision

Understands pictures and video. Used in quality checks and security.

HOW MODELS LEARN

Four ways to train a model

A 'model' is simply the trained result - the file that holds everything the system learned.

1

Supervised

Learn from labelled examples. 'This email is spam.'

2

Unsupervised

No labels. Find natural groups in the data.

3

Reinforcement

Learn by trial, reward and penalty.

4

Self-supervised

Hide part of the data and predict it. Powers Gen AI.

Self-supervised learning is the reason Gen AI arrived so quickly - it can learn from raw internet text without humans labelling it.

03

What Is Generative AI?

From machines that decide to machines that create.

DEFINITION

Generative AI

AI that creates new content - text, images, audio, video or code - from a simple instruction.

- You give it an instruction, called a prompt
- It predicts the most likely next word, pixel or note
- It repeats that prediction until the output is complete
- The result is new, not copied from a database
- Ask twice and you get two different answers

Text

Emails, summaries, reports, code - ChatGPT, Claude, Gemini.

Images and video

Posters, mockups, ads - Midjourney, VEO.

Audio

Voiceovers, music, dubbing in many languages.

The main kinds of Gen AI models

You do not need to build these. You need to know which one to pick.

Large Language Models

Trained on huge amounts of text. Understand and write language. GPT, Claude, Gemini, Llama.

Diffusion models

Start from visual noise and clean it up step by step into an image. Midjourney, Stable Diffusion.

Multimodal models

Accept and produce more than one format - read a chart, answer in words.

Foundation models

Large general models trained once, then adapted cheaply to many specific tasks.

04

Gen AI vs Predictive AI

The comparison that clears up most confusion.

CORE DIFFERENCE

Two different jobs, not two competitors

They work best together, not instead of each other.

PREDICTIVE AI

Answers: what is likely to happen?

- Output is a number, score or category
- Trained on your own historic data
- Answer is stable and repeatable
- Accuracy can be measured exactly
- Example: which customers will leave next month

GENERATIVE AI

Answers: can you create this for me?

- Output is new content
- Trained on very large public datasets
- Answer varies each time you ask
- Quality is judged by a human
- Example: write the offer email to keep them

IN SHORT **Predictive AI tells you what to expect. Generative AI helps you act on it.**

SIDE BY SIDE

Quick reference table

	Predictive AI	Generative AI
Main question	What will happen?	What can be made?
Output	Score, number, label	Text, image, audio, code
Data used	Company's own history	Very large public datasets
Same input twice	Same answer	Different answer
Judged by	Accuracy against reality	Human review of quality
Business use	Forecast demand, score credit	Draft content, summarise, assist
Main weakness	Fails when the future changes	Can state wrong things confidently

If you remember one row, remember the first one.

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05 Advantages, Risks and Limitations

An honest look at both sides.

THE UPSIDE

Why organisations are investing

Speed

First drafts, summaries and reports in minutes instead of days.

Lower cost

Routine work is handled by software, so teams handle more with the same headcount.

Always available

No shifts, no holidays, no drop in quality at 11 pm.

Consistency

The same standard of output every single time.

Scale

Serve ten customers or ten thousand with the same system.

Better decisions

Patterns in large data that a person would simply never spot.

THE DOWNSIDE

Risks you must plan for

None of these are reasons to avoid AI. They are reasons to use it with a process.

Hallucination

- Invents facts, names and figures
- Sounds completely confident while wrong
- Always verify before you use it

Bias

- Learns the bias in its training data
- Can quietly treat groups unfairly
- A real legal and reputation risk

Data leakage

- Staff paste confidential data into public tools
- That data may be stored or reused
- Needs a clear company policy

Over-reliance

- Skills fade when nobody checks the output
- Copy-paste answers with no thought
- Keep a human accountable

LIMITATIONS

What today's AI genuinely cannot do

Knowing the limits is what separates confident users from careless ones.

- **No real understanding**
  It matches patterns in language. It does not know what the words mean.
- **No common sense**
  It can pass a professional exam and then fail a question a child would get right.
- **Knowledge has a cut-off date**
  Unless it is connected to live search, it does not know recent events.
- **Weak at exact maths and logic**
  It predicts likely text, so it can get simple arithmetic wrong.
- **Cannot explain itself**
  It usually cannot show why it produced a particular answer.
- **No accountability**
  A model cannot be responsible. A person always has to be.

USING IT WELL

A simple four-step habit

1

Pick the right task

Drafting, summarising, explaining - not final decisions.

2

Give clear context

Say the role, the goal, the audience and the format.

3

Check the output

Verify every fact, figure, name and quote.

4

Keep the human in charge

A person signs off and owns the result.

Treat AI as a fast junior colleague: helpful, quick, and always reviewed before anything goes out.

SOMETHING TO REMEMBER

AI will not replace you. A person using AI well might.

The skill is no longer doing the task. It is directing the tool and judging the result.

RECAP

Key takeaways

Unit 1 in six lines.

1. **AI learns from data instead of following written rules**
   That is the one difference that explains everything else.

2. **Everything in use today is narrow AI**
   Excellent at one job, useless outside it.

3. **Predictive AI forecasts, Generative AI creates**
   Different jobs. Strongest when used together.

4. **The benefits are speed, cost, scale and consistency**
   Which is why adoption is moving so fast.

5. **The risks are real but manageable**
   Hallucination, bias, data leakage and over-reliance.

6. **A human stays accountable for the output**
   Verify, then approve. Always.

UNIT 01 · INTRODUCTION OF AI AND GEN AI

Thank you

Next up — Unit 2: the platforms, tools and prompt-writing skills that turn these ideas into daily work.

QUESTIONS TO THINK ABOUT

— Which tasks in your own work are drafting rather than deciding?

— Where would a wrong AI answer cost you the most?

— What should your team never paste into a public AI tool?

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